Tangent & Radius

Exam code: 0607

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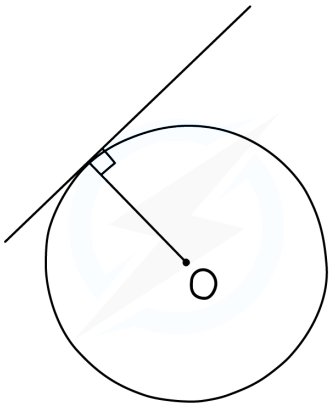
Tangent & Radius

What is a tangent?

- A tangent to a circle is a straight line **outside of the circle** that **touches its circumference only once**

Circle Theorem: A radius and a tangent are perpendicular

- This circle theorem states that a **radius** and a **tangent** meet at **right angles (90°)**
  - This may also be described as being **perpendicular** to each other
- When using this theorem in an exam you must use the **keywords**
  - A **radius** and a **tangent** meet at **right angles** (or 90°)



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Examiner Tips and Tricks

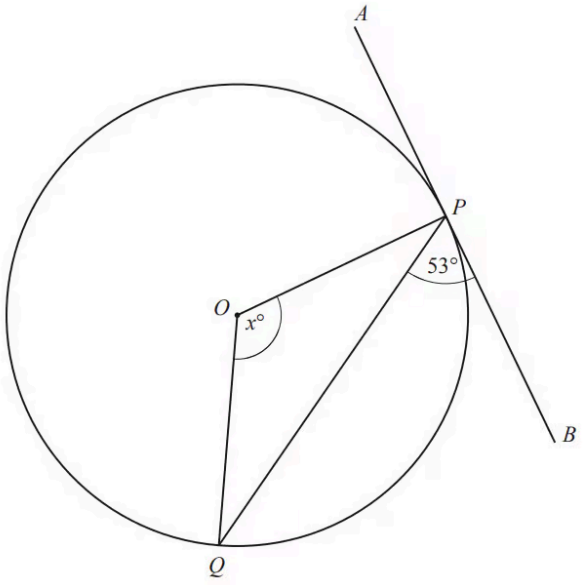
- If you spot a tangent on a circle diagram, look to see if it meets a radius and label the right angle on the diagram
  - In some cases just doing this can earn you a mark!
- If you think you have spotted this circle theorem in a question, make sure it is a **radius** that meets the tangent,

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Worked Example

P and *Q* are points on the circle, centre *O*.

APB is a tangent to the circle at *P*.



(i) Explain why angle *OPB* is 90°.

(ii) Find the value of x.

Answer:

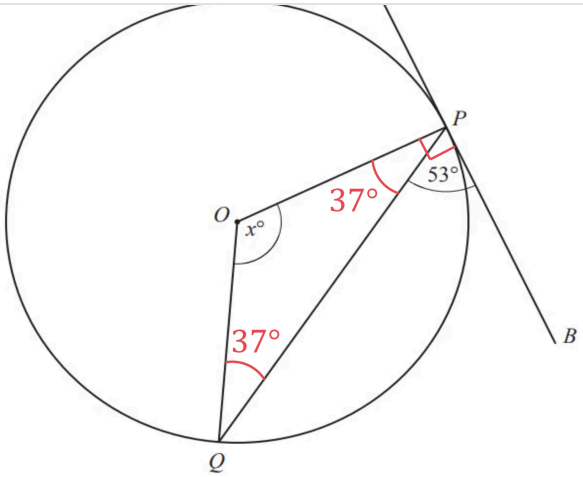
(i)

Angle *OPB* is 90° because the angle between a tangent and a radius is 90° (and *OP* is a radius, and *APB* is a tangent).

(ii) As angle *OPB* is 90°, we can find angle *OPQ*

$$OPQ = 90 - 53 = 37°$$

As *OP* and *OQ* are both the radius of the circle, they have the same length. This means the triangle *OPQ* is isosceles, so the base angles (*OPQ* and *OQP*) are equal



Using the fact that the internal angles in a triangle sum to 180, we can find angle x

$$x + 37 + 37 = 180$$
$$x + 74 = 180$$

$$x = 106$$

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